

Fundamentals of Cyber Defense & SOC Operations

Phase 3 – Threat Detection & Threat Intelligence

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1. CTI Source Classification Table

CTI Source	Type	Intelligence Level	Example Data	How SOC Uses It
CERT-In Advisories	Government	Strategic	Alerts about financial sector attacks	Understand latest threats targeting trading platforms
MITRE ATT&CK	Open Source	Operational	Attack techniques database	Map detected attacks to known adversary behavior
VirusTotal	Open Source	Tactical	Malicious file hashes, IP reputation	Check suspicious files from trading servers
Security Vendor Reports (CrowdStrike, Palo Alto)	Commercial	Strategic	Threat trends and APT reports	Improve SOC detection strategy
Internal SOC Alerts	Internal	Tactical	SIEM alerts and incident reports	Detect repeated attack patterns
Firewall Threat Intelligence Feed	Commercial	Operational	Malicious IP lists	Block attacker infrastructure
Dark Web Monitoring Reports	Commercial	Strategic	Leaked credentials	Detect possible insider compromise

2. IoC and IoA Mapping

Selected Attack Scenario:

Unauthorized Trading Account Access (Brute Force Attack)

IoC vs IoA Explanation:

IoC (Indicator of Compromise)

Evidence that a system may already be compromised.

Examples:

- Malicious IP
- Suspicious login logs
- Modified files

IoA (Indicator of Attack)

Attacker behavior pattern.

Examples:

- Repeated login attempts
- Privilege escalation
- Unauthorized execution

IoC & IoA Table

Attack Stage	IoC	IoA	Log Source	Why It Matters
Initial Access	Multiple failed login events	Password brute force attempts	Windows Security Logs	Detect credential attacks
Initial Access	Login from foreign IP	Unusual login location	Firewall logs	Detect unauthorized access
Execution	Suspicious script execution	Running unknown process	EDR logs	Detect malware execution
Persistence	New admin account created	Privilege escalation attempt	Active Directory logs	Detect attacker persistence
Defense Evasion	Log deletion activity	Attempt to hide activity	SIEM logs	Detect attacker cleanup actions

Minimum IoCs identified:

- Malicious IP address
- Failed login events
- Unauthorized account creation
- Suspicious process hash
- Log tampering activity

Minimum IoAs identified:

- Repeated login attempts
- Privilege escalation behavior
- Suspicious process execution

3. MITRE ATT&CK Technique Mapping**Attack Scenario:**

Unauthorized trader account compromise leading to algorithm manipulation.

MITRE Mapping Table

Attack Step	MITRE Tactic	Technique ID	Technique Name	Description
Phishing email sent	Initial Access	T1566	Phishing	Attacker tricks trader into revealing credentials
Malicious script execution	Execution	T1059	Command Execution	Attacker runs scripts on workstation
Credential stealing	Credential Access	T1003	Credential Dumping	Attacker extracts passwords
Create persistence	Persistence	T1078	Valid Accounts	Uses stolen accounts to maintain access
Modify logs	Defense Evasion	T1070	Indicator Removal	Deletes logs to hide activity
Modify trading algorithm	Impact	T1565	Data Manipulation	Changes algorithm behavior

MITRE ATT&CK Technique Mapping Explanation

In our project, we considered an attack scenario where an attacker gains unauthorized access to the HFT trading environment and tries to manipulate trading systems. Since our organization depends on automated trading algorithms and real-time market data, any unauthorized access can cause financial loss and market instability. To understand how such attacks happen, we mapped the attack steps to the MITRE ATT&CK framework.

Initial Access – Phishing (T1566)

In this step, the attacker may send phishing emails to traders or developers working in the HFT firm. Since traders regularly receive financial reports and trading updates, attackers may

disguise emails as market reports. If the trader clicks the malicious link and enters credentials, the attacker can gain access to the trading workstation. This is usually the starting point of the attack.

Execution – Command and Scripting Interpreter (T1059)

After gaining access, the attacker may execute commands or scripts on the compromised trader workstation or trading server. For example, they may run scripts to collect system information or install malware. In our HFT environment, this could allow attackers to understand how trading algorithms work.

Credential Access – Credential Dumping (T1003)

Once inside the system, the attacker may try to collect more credentials from the compromised machine. They may attempt to extract stored passwords or authentication tokens. This could allow them to access critical systems such as the order execution engine or risk management server.

Persistence – Valid Accounts (T1078)

To maintain access, attackers may continue using the stolen trader account or create a new unauthorized account. Since they are using legitimate credentials, their activity may look normal. In our project, this could allow attackers to continuously access trading systems without immediate detection.

Defense Evasion – Indicator Removal (T1070)

Attackers may try to delete logs or modify records to hide their activities. Since our SOC depends on logs from trading servers, SIEM, and firewalls (as discussed in Phase-2), log deletion could delay detection. This makes the attack more dangerous because SOC analysts may not immediately notice the compromise.

Impact – Data Manipulation (T1565)

In the final stage, the attacker may try to modify trading algorithm parameters or market data. This could cause incorrect trading decisions or abnormal trading behavior like flash crash situations (identified as a risk in Phase-1). This stage directly impacts trading integrity and financial stability.

4. Threat Hunting Hypotheses

Hypothesis 1

Hunting Hypothesis:

If a trading algorithm file is modified during active trading hours without change approval, it may indicate insider manipulation or attacker activity.

Data Sources to Check:

- File integrity monitoring logs
- Trading server syslogs

- Application logs
- SIEM alerts

Why This Is Suspicious:

Trading algorithms normally change only during maintenance windows. Unauthorized changes may indicate manipulation.

Expected Outcome:

Detection of unauthorized modifications or insider misuse.

Hypothesis 2**Hunting Hypothesis:**

If multiple trader login failures occur followed by a successful login from a new device, it may indicate account compromise.

Data Sources:

- Windows login logs
- Active Directory logs
- Firewall logs
- SIEM authentication alerts

Why Suspicious:

This pattern matches brute force attack behavior.

Expected Outcome:

Identification of compromised accounts.

Hypothesis 3**Hunting Hypothesis:**

If trading server latency suddenly increases along with abnormal network traffic, it may indicate DoS attack activity.

Data Sources:

- Network logs
- Firewall logs
- Trading performance logs
- SIEM alerts

Why Suspicious:

HFT systems depend on low latency. Any delay may indicate attack attempts.

Expected Outcome:

Detection of availability attacks on trading infrastructure.